

LITHO Mainnet 9005 Ownership and Admin-Control Handover

Last reviewed: 2026-07-29

Accountable organization: **KaJ Labs**. Individual operators, custodians, Safe signers, and escalation contacts remain to be named in the client-controlled operating record. An organization or Safe address does not replace the need for people who can respond to an incident.

This document defines the operational ownership and technical controls covered by the handoff. It does not disclose credentials, private keys, recovery material, or secret-manager values.

1. Control model

The handoff uses three separate concepts:

- 1. **Accountable owner:** KaJ Labs is responsible for approving changes, accepting risk, and ensuring the control remains operated.
- 2. **Operational custodian:** an authorized person or team performs recovery, monitoring, deployment, or key-custody procedures.
- 3. **Technical control:** the actual Safe, key, provider account, IAM role, DNS account, SSH credential, or service configuration that authorizes an action.

A Safe multisig is appropriate for compatible on-chain treasury and contract administration. It cannot directly replace a CometBFT consensus key, an SSH key, a VPS-provider login, a DNS/Cloudflare account, or an on-call owner.

2. Ownership and control matrix

Area	Accountable owner	Control handed over or confirmed	Handoff state	Required client action
Mainnet service acceptance and risk decisions	KaJ Labs	Acceptance checklist and evidence package	Delivered for conditional acceptance	Select the handoff decision and record target dates
Source repository and deployment automation	KaJ Labs	Repository access and documented Ansible/scripts	Access confirmation pending	Confirm authorized client administrators and branch/release policy
Validator VPS administration	KaJ Labs	VPS-provider control plus key-only SSH access	Client access must be confirmed	Retain two independent administrators and remove obsolete access after validation
Sentry VPS administration	KaJ Labs	VPS-provider control plus key-only SSH access	Client access must be confirmed	Validate access to both sentries and preserve one-at-a-time maintenance
Validator consensus signing	KaJ Labs	Encrypted offline identity copies and the live validator service	Initial copies confirmed; recurring current-state backup remains open	Name two custodians and implement/test current signing-state backup

Area	Accountable owner	Control handed over or confirmed	Handoff state	Required client action
Validator operator account	KaJ Labs	On-chain validator operator account and its signing authority	Custody must be confirmed through the approved secure channel	Test a non-destructive query/signing procedure; do not expose the key in chat or the repository
Validator signing state	KaJ Labs	<code>priv_validator_state.json</code> , paired with the active consensus key	Live state exists; recurring protected backup remains open	Preserve current height/round/step and enforce single-writer recovery
P2P and private validator mesh	KaJ Labs	<code>wg-mainnet</code> topology and node configuration	Running	Maintain WireGuard and P2P access without exposing validator APIs publicly
Public RPC, REST, gRPC, and WebSocket	KaJ Labs	TLS proxy and sentry node services	Live	Monitor availability, certificate renewal, rate limits, and consumer migration from raw ports
DNS, Cloudflare, and certificates	KaJ Labs	DNS/proxy configuration and Certbot renewal path	Client account access confirmation pending	Name administrators, alert on expiry, and test renewal/reload periodically
Monitoring and incident response	KaJ Labs	Runbook, health checks, and verification script	Mainnet-specific paging remains open	Assign on-call coverage and test the no-block alert and escalation path
Lithoscan explorer	KaJ Labs	Explorer code, production deployment, Cloudflare routing, restricted deployment identity, and rollback procedure	Live; cutover and monitoring closeout passed 2026-07-31	Maintain monitoring, certificate/edge ownership, release controls, and tested rollback capability
Genesis and network identity	KaJ Labs	Checksum-pinned genesis, chain IDs, binary hash, and launch evidence	Delivered and immutable	Archive verified copies; do not regenerate or edit genesis
Native allocation and treasury wallets	KaJ Labs	Client-supplied allocation addresses	No private custody material is contained in the handoff repository	Confirm custody and decide whether eligible EVM treasury controls move to a tested Safe
Governance and economic decisions	KaJ Labs	Current genesis parameters and documented open decisions	Owner assigned; final policy record pending	Record governance owner, allocation labels, and fee/burn policy
LEP100 token administration	KaJ Labs or the applicable token issuer	No mainnet LEP100 registry or production token-admin key is represented as delivered	Outside completed base-chain handoff	Publish an approved registry and per-token owner/Safe policy before exchange listing
Bridge, MultX, Swap, and Faucet	KaJ Labs	No production admin control; these components are disabled	Outside initial mainnet scope	Complete separate deployment, audit, governance, and custody acceptance before activation

3. Can KaJ Labs change the validator signing key after handoff?

Yes, KaJ Labs can initiate and control a validator-key replacement after handoff, but the active validator's consensus key must not be changed by simply replacing `priv_validator_key.json`. The consensus public key is part of the on-chain validator identity. An uncoordinated file replacement will not rotate that identity and can stop block production. Reusing the old key on two hosts can cause double-signing and permanent validator penalties.

For this Cosmos SDK/CometBFT validator, treat consensus-key change as a coordinated validator replacement:

1. KaJ Labs approves a maintenance plan, rollback conditions, and named key custodians.
2. Generate the new operator and consensus keys in the client-approved custody environment. Never generate or transmit them in chat or version control.
3. Back up and checksum the current `priv_validator_state.json`; prove that the existing validator remains the only process using the old consensus key.
4. Provision and fully synchronize a new validator host without starting it as an unauthorized replacement signer.
5. Create the new on-chain validator identity with transferred existing LITHO stake. The fixed supply means no new stake may be minted for the rotation.
6. Coordinate voting-power movement so the chain retains more than two-thirds active voting power throughout the transition. This is critical while the network has only one active validator.
7. Verify the new validator is bonded and signing, then retire/unbond the old validator under an approved sequence.
8. Observe block production and all public endpoints, archive the change record, and revoke or quarantine the old key according to the custody policy.

The exact transaction sequence, stake amounts, and timing must be rehearsed on a matching non-production network and approved for the live validator set. This is a chain-level maintenance operation, not a normal server key rotation.

4. Controls that can be rotated independently

Control	Can KaJ Labs change it after handoff?	Important condition
Authorized SSH public keys	Yes	Test replacement access before removing the prior key
VPS-provider users and credentials	Yes	Keep two independent administrators and provider recovery access
DNS/Cloudflare administrators and API tokens	Yes	Preserve rollback and avoid changing chain endpoints without notice
TLS certificates	Yes	Certbot is staged; monitor expiry and validate Nginx reload
WireGuard keys	Yes	Rotate one peer at a time and preserve validator-to-sentry connectivity
CometBFT node identity key	Yes, with coordination	Peer identity changes; update persistent peers and address books as required
Validator consensus key	Not in place	Use the coordinated validator-replacement procedure in section 3
Validator operator-account key	Not in place	An account key defines its address; move control through a planned validator replacement rather than overwriting it
Safe signers and threshold	Yes, for a deployed and tested Safe	Follow the Safe policy and verify the chain-9005 deployment before relying on it

Control	Can KaJ Labs change it after handoff?	Important condition
Smart-contract owner/admin	Yes, when a contract supports transfer	Verify the exact deployed address and execute a two-step or timelocked transfer where available
Genesis, chain ID, or genesis allocations	No	These are immutable network identity; changing them creates a different chain

5. Secure handover evidence

The client acceptance record should reference evidence, not secret values:

- KaJ Labs account administrators can access the repository, VPS provider, DNS/Cloudflare, and monitoring systems;
- two authorized operators can independently access the nodes;
- validator identity backups remain encrypted, separately stored, and recoverable;
- the current signing-state backup and restore drill have named custodians;
- Safe addresses, signer threshold, and signer identities are recorded in the client-controlled custody register when applicable;
- obsolete delivery-team access is revoked only after replacement access is tested; and
- no key, seed phrase, password, or recovery material is placed in the signed checklist, shared group, Google Doc, or repository.

6. Acceptance boundary

This document assigns KaJ Labs as the accountable organization. It does not claim that every technical-control transfer is already complete. Rows marked pending must be validated in the client-controlled acceptance checklist, and the remaining Critical and High operational gates require closure or explicit risk acceptance.

7. Technical references

- The deployed Evmos `v20.0.0` release uses Cosmos SDK `v0.50.9` and CometBFT `v0.38.12`: [Evmos v20.0.0 release](#).
- In Cosmos SDK `v0.50.9`, `MsgEditValidator` changes validator description, commission rate, and minimum self-delegation; it does not accept a new consensus public key: [Cosmos SDK staking message reference](#).
- CometBFT describes the consensus key as the key used by a validator to sign consensus votes and emphasizes protected key handling: [CometBFT validator documentation](#).